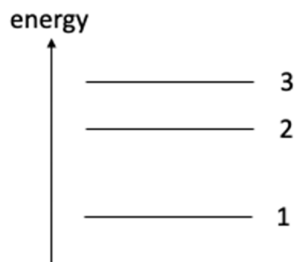


- 27 Part of the energy level diagram of a certain atom is shown below. The energy spacing between levels 1 and 2 is three times that between 2 and 3. If an electron makes a transition from level 3 to level 2, the radiation of wavelength λ is emitted.



What possible radiation wavelengths might be produced by other transitions between the three energy levels?

A only $\frac{\lambda}{3}$

B both $\frac{\lambda}{3}$ and $\frac{\lambda}{4}$

C only 3λ

D only 4λ

Answer: B

$$E_{3 \rightarrow 2} = \frac{hc}{\lambda}$$

Possible transitions:

$$\begin{aligned} E_{2 \rightarrow 1} &= 3E_{3 \rightarrow 2} \\ &= 3 \frac{hc}{\lambda} \\ &= \frac{hc}{(\lambda/3)} \end{aligned}$$

$$\begin{aligned} E_{3 \rightarrow 1} &= 4E_{3 \rightarrow 2} \\ &= 4 \frac{hc}{\lambda} \\ &= \frac{hc}{(\lambda/4)} \end{aligned}$$

Therefore, these transitions will produce the wavelengths $\frac{\lambda}{3}$ and $\frac{\lambda}{4}$.